OEAS 705/805 Advanced Environmental Data Science

Fall 2026

October 20, 2026

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Schedule

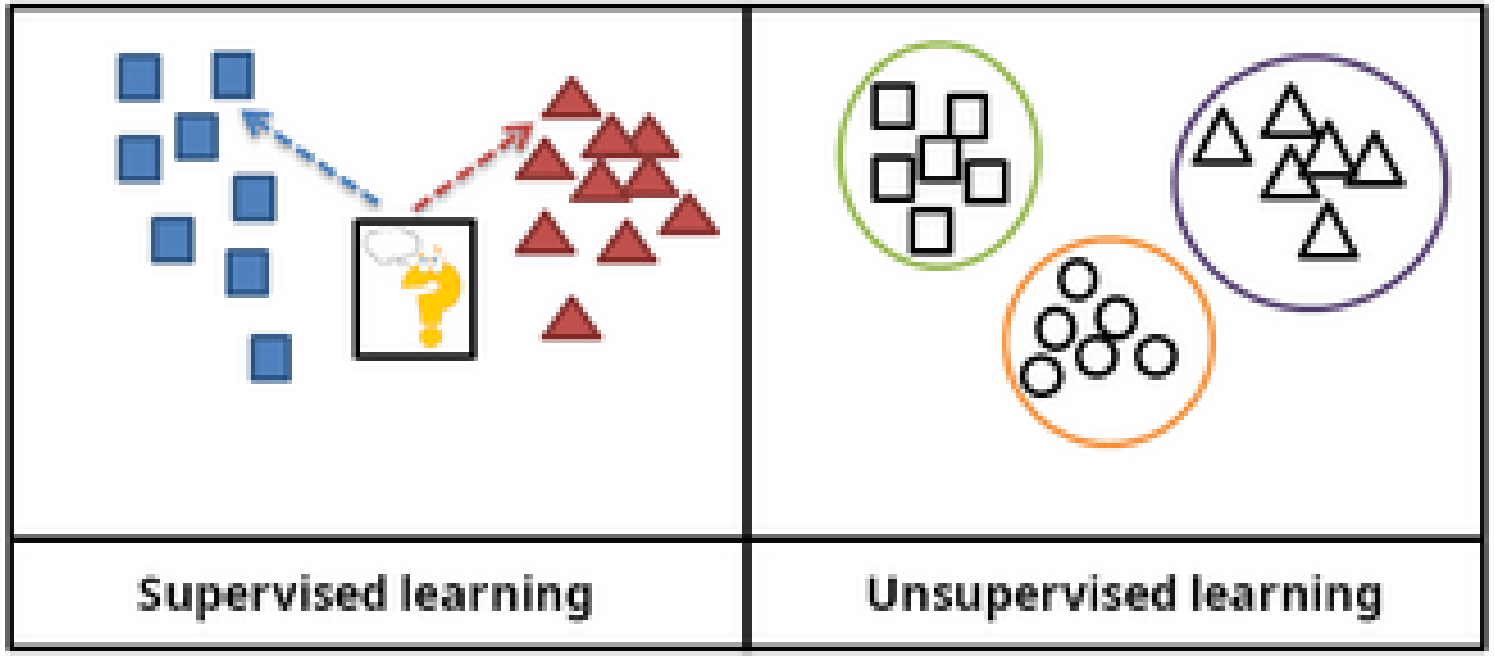
- HW 4 will be posted soon - Due Monday November 2nd.
- No class Tuesday November 3rd for Election Day.
- Projects complete November 23rd for Peer Review
- Peer Reviews Due by November 30th (graded for completion only)

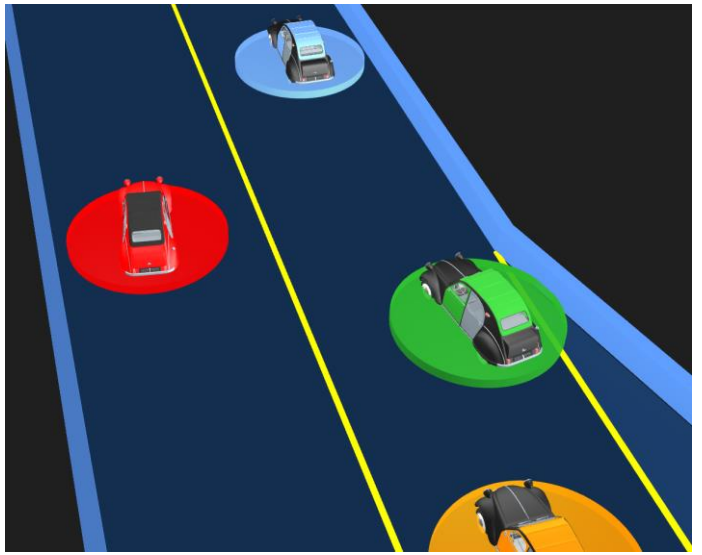
Machine Learning

1. Supervised Learning

2. Unsupervised Learning

3. Reinforcement Learning

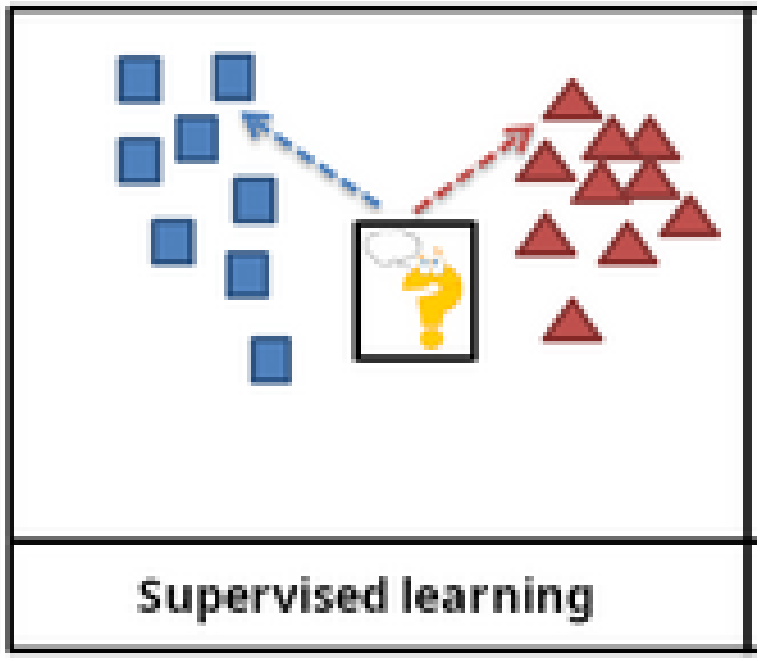




Game Theory applications,
Not as useful in earth sciences

Machine Learning

1. Supervised Learning



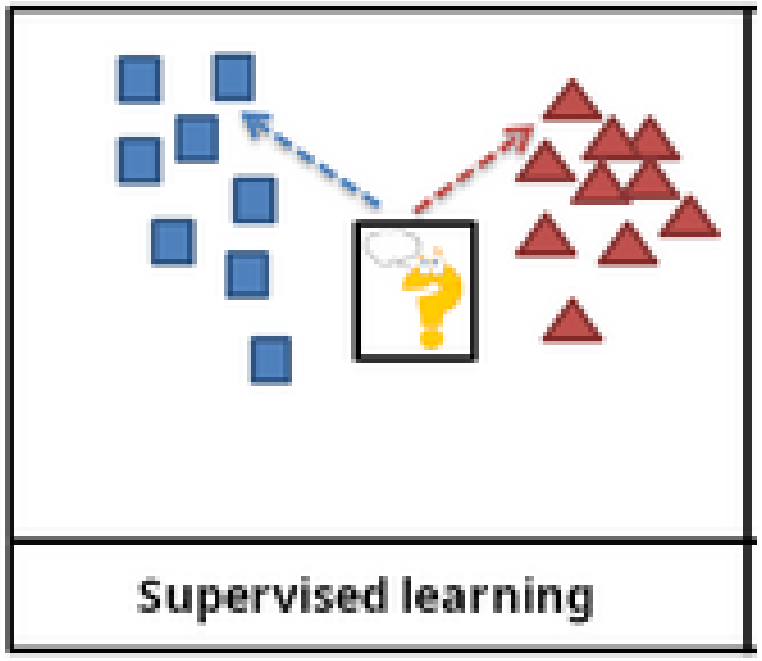
Labeled samples are used to train the model

- Regression – predict values
- Classification – predict category

Discuss: what do you think is more common in Earth/Environmental Sciences right now (2026)?

Machine Learning

1. Supervised Learning

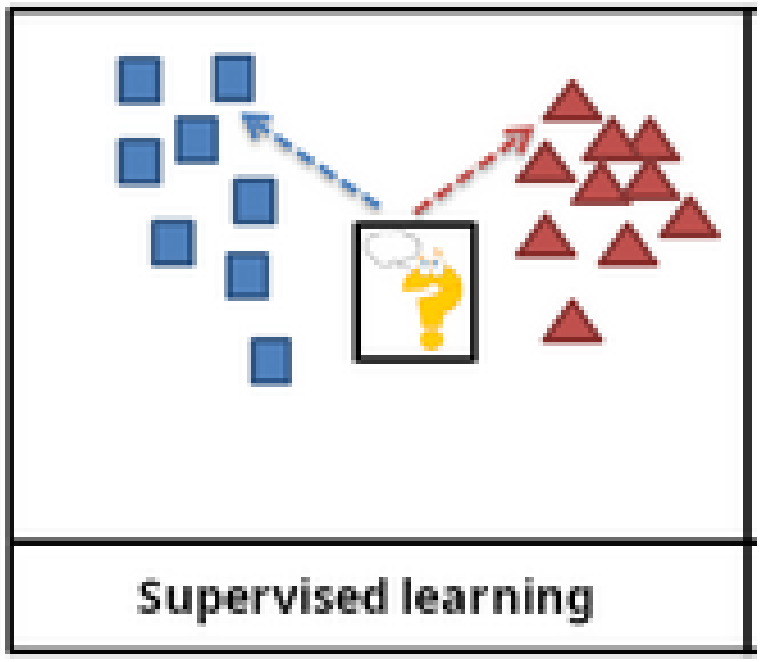


Labeled samples are used to train the model

- Regression – predict values
- **Classification – predict category**

Machine Learning

1. Supervised Learning



Classification Examples:

- Support Vector Machines
- Random Forest
- K-Nearest Neighbors

Yes, you can use them for regression OR classification…

Supervised ML Models for *Classification*

Some applications:

- **Mapping land cover:** "Forested" vs. "Urban" vs. ”Agricultural” etc.
- **Classifying coral health:** "Bleached" vs. "Healthy”
- **Detecting benthic plant type:** “Cladophora” vs. “Eelgrass” vs. “Sand”
- **Merlin Bird ID app:** different types of birds based on their song or picture.

Supervised ML Models for *Classification*



Supervised ML Models for *Classification* that we will try out:

- K-Nearest Neighbors
  classification_iris_KNN_example.ipynb

- Multi-Layer Perceptron Neural Network
  classification_iris_neuralnet_example.ipynb